DDD–T–541c
July 23, 1965
SUPERSEDING
Fed. Spec. DDD-T-541b
December 4, 1961

FEDERAL SPECIFICATION

TOWEL, MACHINERY WIPING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements of new cotton wiping towels which can be reused after laundering.

1.2 Classification. Towels shall be of the following sizes (see 6.2):

Size 1—16-1/2 by 18 inches.
Size 2—18 by 30 inches.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276—Thread, Cotton.
UU-P-268—Paper, Kraft, Untreated, Wrapping.
CCC-T-191—Textile Test Methods.
DDD-L-20—Label; For Clothing, Equipage, and Tentage (General Use).
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751—Stitches, Seams, and Stitching.
Fed. Std. No. 4—Glossary of Fabric Imperfections, Section 1.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Federal Trade Commission (FTC) Publication:

Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C.)

3. REQUIREMENTS

3.1 Sample. When a sample is available, it will be furnished as a guide only. Variations from this specification may appear in the sample, in which case this specification shall govern (see 6.4).

3.2 Preproduction sample. When specified (see 6.2), before production is commenced, a preproduction sample of two or more finished towels shall be submitted or made ready for approval of the contracting officer or his authorized representative for inspection and tests as specified in section 4 to determine compliance with this specification. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification.

3.3 Materials.

3.3.1 *Cotton.* The cotton shall be of a suitable length of staple and grade, and shall be spun in single ply yarn to conform to this specification.

3.3.2 *Weave.* The weave of the towels shall be plain.

3.3.3 *Physical requirements.* The material shall comply with the minimum requirements of table I.

TABLE I. *Physical requirements*

Weight, ounces per square yard (min.)	Yarns per inch (min.)		Breaking strength, pounds (min.)	
	Warp	*Filling*	*Warp*	*Filling*
5.5	28	18	40	75

3.3.4 *Nonfibrous material content.* The starch and protein content including chloroform-soluble and water-soluble material of the finished towels shall not exceed 7.0 percent when tested as specified in 4.5.

3.3.5 *Finish.* The material may be bleached or unbleached as specified (see 6.2).

3.4 Dimensions. Towels shall be made of one piece of material. The finished dimensions of size 1 towel shall be not less than 16-1/2 inches wide and not less than 18 inches long. Size 2 towels shall measure not less than 18 inches wide and not less than 30 inches long.

3.5 Labeling. Labeling shall be in accordance with type III or IV class 5 of DDD-L-20 for laundered items. The marking shall be distinct with edges of all letters sharp and clear cut and shall bear "PROPERTY OF U. S. GOVERNMENT" in solid block letters 1 inch high and 3/8 inch wide, applied in the approximate center of each towel.

3.6 Fiber identification. Each towel shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.7 Hems and selvages. All cut edges of the towel, except for selvage on one side, shall be overedged with stitch type 503, or hemmed 5/32 (± 1/32) inch with stitching type 301 (EFb-1) of Fed. Std. No. 751 and shall have not less than eight stitches per inch. There shall be no raw edges.

3.8 Thread. The thread used shall not be finer than ticket No. 30, 3-ply, soft finish, unbleached cotton conforming to type 1A1 of V-T-276.

3.9 Workmanship. The towels shall be clean and free from defects which affect the appearance or which may impair the serviceability, and shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. The preproduction sample submitted in accordance with 3.2, shall be inspected to verify compliance with this specification.

4.2.1 *Inspection of components and materials.* Quality assurance provisions for components and materials described, shall be in accordance with this specification or subsidiary specifications referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.3 Inspection for acceptance. Sampling and inspection shall be performed in accordance with MIL-STD-105, except as otherwise specified hereinafter.

4.3.1 *Examination of the end item.*

4.3.1.1 *Visual examination.* Defects found during this examination shall be classified in accordance with Fed. Std. No. 4 and the defects listed below. All defects shall be counted regardless of their proximity to each other, except when two or more defects represent a single local condition in the item, in which case only the more serious defect shall be counted. The unit of product for this examination shall be one finished towel. The maximum number of sample units to be drawn from any one shipping container shall be 20. The acceptable quality level (AQL) for this examination shall be 2.5 major and 6.5 total defects (major and minor combined) per 100 units. The sample size shall be in accordance with inspection level I of MIL-STD-105. The lot size shall be expressed in units of one finished towel. Defects shall be classified as follows:

Examine	Defect	Classification	
		Major	Minor
Material	Cut, hole or tear ...	X	
	Smash	X	
	Floats and skips:		
	Multiple 1 inch or in combined directions	X	
	Multiple more than 1/4 inch but less than 1 inch in combined directions		X
	Single 4 inches or more in length ..		X
	Broken or missing yarns:		
	Two or more contiguous	X	
	Single 4 inches or more in length ..		X
	Fine or light filling bar, (a light place)		X
	Hitchback, crack, or open place:		
	Opening between yarns 1/16 inch or more at widest place		X
	Sequence of openings resulting from slack warp ends resulting in a weak spot		X
	Slub, slug or gout, jerked-in filling or slough off more than 1/8 inch in thickness		X
	Spot, stain or streak clearly visible at normal inspection distance		X
	Selvage (when applicable, on the side):		
	Cut, torn, broken or omitted		X

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Examine	Defect	Classification	
		Major	Minor
Material (cont'd)	Scalloped or permanently curled .		X
	Not bleached (when bleached as specified, see 6.2)		X
	Objectionable odor .		X
Workmanship	Hem omitted, except when selvage along one side is applicable	X	
	Hem less than 1/8 inch or more than 3/16 inch wide ..		X
	Stitching less than 8 stitches per inch (when applicable)	X	
	Raw edge, or overedge stitching missing, 3/4 inch or more	X	
	Raw edge or overedge stitching missing up to 3/4 inch		X
	Wrong stitch type .		X
	Needle chew likely to develop into a hole	X	
	Skipped, broken or run-off stitching, 3/4 inch or more .	X	
	Skipped, broken or run-off stitching, up to 3/4 inch ...		X
Identification marking	Government ownership marking omitted, incomplete, illegible, incorrect wrong color		X

4.3.1.2 *Dimensional examination.* Any dimension below the minimum listed below shall be classified as a defect. The unit of product shall be one finished towel. The number of containers examined shall be those sampled for visual examination (see 4.3.1.1). The AQL for this examination shall be 4.0 defects (one class) per 100 units. The sample size shall be in accordance with inspection level S-3, of MIL-STD-105. The lot size shall be expressed in units of one finished towel.

Overall width:

Size 1 towels—less than 16-1/2 inches.
Size 2 towels—less than 18 inches.

Overall length:

Size 1 towels—less than 18 inches.
Size 2 towels—less than 30 inches.

4.4 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and marking requirements of section 5 are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packed, selected prior to the closing operation. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing or not as specified. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Content	Number of bundles per container is more or less than specified. Number of towels per bundle is more or less than specified.[1]

[1] For this defect, 3 bundles shall be examined from each shipping container in the sample.

4.5 Testing of the end item. The methods of testing specified in CCC-T-191 wherever applicable and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the average of determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be

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two towels. The sample size shall be in accordance with table II. The lot size shall be expressed in units of 1 towel each. The lot shall be unacceptable if one or more sample units fail to meet a test requirement.

TABLE II. *Sampling for end item testing*

Lot size	Sample size (sample units)
800 or less	2
801 to 22,000	3
22,001 and up	5

TABLE III. *Instructions for testing*

Characteristic	Requirement paragraph	Test method	Results reported as
Material identification	3.3.1	1200	Pass or fail
Weave	3.3.2	Visual[1]	Pass or fail
Weight, ounces per square yard	Table I	5041	Pass or fail
Yarns per inch	Table I	5050	Pass or fail
Breaking strength	Table I	5100	Pass or fail
Nonfibrous material	3.3.4	2611	To nearest 0.1 percent

[1] One determination per sample unit shall be taken for this characteristic.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 should be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 Packaging. Packaging shall be level A or C (see 6.2).

5.1.1 *Level A.* Fifty size 1 towels shall be evenly stacked one on top of the other. Fifty size 2 towels shall be evenly stacked one on top of the other with the thirty inch dimension folded in half. Each bundle of 50 towels shall be tied across the center with cloth tape or soft cotton twine to form a bundle.

5.1.2 *Level B (civil agencies).* The towels shall be packaged as specified in 5.1.1.

5.1.3 *Level C.* Towels shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C (see 6.2).

5.2.1 *Level A.* Five hundred size 1 towels (10 bundles of 50 towels each); or two hundred and fifty size 2 towels (5 bundles of 50 towels each), of one size only, packaged as specified in 5.1.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V3c or V3s of PPP-B-636. The top and bottom of each pack of towels shall be completely covered with a sheet of kraft paper of not less than 30 pounds basis weight conforming to grade B of UU-P-268. Each shipping container shall be closed, waterproofed and reinforced with flat steel strapping or tape banding in accordance with appendix of the container specification.

5.2.2 *Level B.* Five hundred size 1 towels (10 bundles of 50 towels each); or two hundred and fifty size 2 towels (5 bundles of 50 towels each), of one size only, packaged as specified in 5.1.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF or SF, class-domestic, variety SW, grade 275 of PPP-B-636. The top and bottom of each pack of towels shall be completely covered with a sheet of kraft paper of not less than 30 pounds basis weight conforming to grade B of UU-P-268. Each shipping container shall be closed in accordance with method II of the appendix of the container specification.

5.2.3 *Level C.* Towels shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Shipping containers shall conform to carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The towels are primarily intended for use in cleaning machinery and mechanical components.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size of towels required (see 1.2).
(*c*) When preproduction sample is required (see 3.2).
(*d*) Whether bleached or unbleached finish is required (see 3.3.4).
(*e*) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.3 Transportation description. The transportation description and minimum weight applicable to this commodity are;

Rail:

Towels, cotton
Carload minimum weight—any quantity

Motor:

Dry goods, not otherwise indexed
Motor volume minimum weight—any quantity

6.4 Samples. For access to samples of towels address the procuring office issuing the invitation for bids.

MILITARY CUSTODIANS:

Army—GL
Navy—SA
Air Force—69

Interested activities:

Review

Army—GL, MD
Navy—SA
Air Force—69

Preparing Activity:

Navy—SA (NAVSUPRANDFAC)

User

Navy—BUDOCKS, BUSHIPS, MARCORPS

Reviewer and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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Copies of this specification may be purchased for cents each.

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